

SIMATS ENGINEERING

CO3 – ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

Course: Digital Signal Processing (ECA09)

CO3: Finite Word-Length Effects and Multirate Signal Processing

Question: A DSP processor with Q15 representation overflows while multiplying 0.95×0.92 . Determine the cause and suggest scaling corrections.

1. Given Data

- DSP representation: Q15 fixed-point
- First input: $x = 0.95$
- Second input: $y = 0.92$
- Required product: 0.95×0.92

$$0.95 \times 0.92 = 0.874$$

Therefore, the mathematical result is within the Q15 range.

2. Q15 Representation

For signed Q15, there is 1 sign/integer bit and 15 fractional bits. The conversion from a real number to its Q15 integer representation is:

$$X_{Q15} = x \times 2^{15}$$

The representable range is approximately:

$$-1 \leq x \leq 1 - 2^{-15} = 0.999969$$

Hence both 0.95 and 0.92 are valid Q15 values.

$$0.95 \times 32768 = 31129.6 \approx 31130$$

$$0.92 \times 32768 = 30146.56 \approx 30147$$

Therefore: $0.95 \rightarrow 31130$ and $0.92 \rightarrow 30147$

3. Multiplication and the Implementation Error

When two Q15 numbers are multiplied, their fractional bits add:

$$Q15 \times Q15 = Q30$$

Thus the processor first produces a Q30 product. Using the Q15 integer values:

$$31130 \times 30147 = 938,476,110$$

This value is a Q30 result. If the processor incorrectly attempts to store this full product directly in a 16-bit Q15 register, the register cannot hold it and overflow occurs.

Root cause: The Q30 multiplication result is being treated as a Q15 result without the required rescaling.

4. Correct Scaling After Multiplication

To convert the Q30 product back to Q15, the product must be shifted right by 15 bits:

$$PQ15 = (XQ15 \times YQ15) / 2^{15}$$

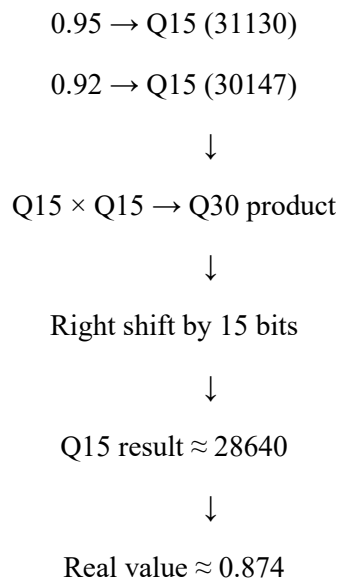
$$PQ15 = 938,476,110 / 32768 \approx 28640$$

$$\text{Real product} \approx 28640 / 32768 = 0.8740$$

Correct result: $0.95 \times 0.92 \approx 0.874$, with no overflow when the multiplication is implemented with proper scaling.

5. Corrected DSP Implementation

The correct fixed-point multiplication flow is:



6. Signal Scaling / Headroom Correction

In a DSP system containing several additions or multiplications, additional headroom may be required. A scaling factor α , where $0 < \alpha < 1$, can be applied before processing:

For example, if $\alpha = 0.5$:

$$0.95 \times 0.5 = 0.475$$

$$0.92 \times 0.5 = 0.46$$

$$0.475 \times 0.46 = 0.2185$$

This provides extra numerical headroom and reduces the possibility of overflow in later processing stages. The scaling factor must be accounted for when restoring the desired output amplitude.

7. Saturation Arithmetic

Proper scaling should be combined with saturation arithmetic. If a Q15 result exceeds its allowed range, it should be limited rather than allowed to wrap around.

- Maximum positive Q15 value $\approx +0.999969$
- Minimum negative Q15 value $= -1$

Saturation prevents an overflow from wrapping into an incorrect sign and producing a severely distorted DSP output.

8. Error / Root-Cause Analysis

Implementation issue	Root cause	Correction
Q15 multiplication overflow	Q30 product stored directly as Q15	Use a wider product/accumulator
Incorrect output magnitude	Missing 15-bit rescaling	Right shift product by 15 bits
Overflow in later DSP stages	Insufficient signal headroom	Apply suitable signal scaling
Wrap-around distortion	No overflow protection	Use saturation arithmetic

9. Corrected Calculation Summary

$$0.95 \times 0.92 = 0.874$$

$$0.95 \rightarrow 31130 \text{ (Q15)}$$

$$0.92 \rightarrow 30147 \text{ (Q15)}$$

$$31130 \times 30147 = 938,476,110 \text{ (Q30)}$$

$$\text{Q30} \rightarrow \text{Q15: } 938,476,110 \gg 15 \approx 28640$$

$$28640 / 32768 \approx 0.8740$$

10. Conclusion

The overflow is not caused by the mathematical product 0.95×0.92 because the result, 0.874, lies within the Q15 range. The implementation error occurs because multiplying two Q15 numbers produces a Q30 result, which cannot be directly stored in a 16-bit Q15 register. The correct solution is to use a wider accumulator for the multiplication and shift the Q30 result right by 15 bits before storing it as Q15. For larger DSP processing chains, suitable signal scaling should be used to provide headroom, together with saturation arithmetic to prevent wrap-around. These corrections minimize finite-word-length errors and improve the reliability of fixed-point DSP implementations.

Key correction: $\text{Q15} \times \text{Q15} \rightarrow \text{Q30} \rightarrow \text{shift right 15 bits} \rightarrow \text{Q15}$